

Volume 5 — Maintenance & Troubleshooting

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

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Chapter 1 — Maintenance Philosophy & Cross-Reference

The machine's mechanics are unchanged by the retrofit, so the **original Mazak Maintenance Manual (413m033)** remains the authority for ways, ballscrews, spindle bearings, hydraulics, and lubrication intervals. This volume covers what the original manual cannot: the Mesa/LinuxCNC control layer, and the boundary between old iron and new electronics.

Topic	Authority
Mechanical maintenance, lube points, hydraulic specs	413m033 (Maintenance Manual)
Machine electrical (motors, transformer, contactors)	41434WB (Electrical Circuit Diagram)
Original interlock logic / signal meanings	YM2V39L (PLC Ladders) + element list CSV
Spare mechanical parts	413LE02A000 (Parts List)
Control layer (PC, Mesa, HAL, wiring to Mesa)	This manual set (Vols 3–4)

Chapter 2 — Daily / Weekly / Monthly Schedules

Daily (operator):

- Check way-lube reservoir; verify `lube-level` shows OK.
- Check hydraulic pressure gauge and `hydraulic-ok` state.
- Check coolant level and condition.
- Listen during first warm-up: spindle, hydraulic pump, ATC noises.
- Clear chips from way covers, ATC arm area, and door tracks.

Weekly:

- Grease/oil points per 413m033 schedule.
- Inspect resolver and analog cable runs for chafing or oil intrusion.

- Verify E-stop chain: press each button; confirm contactors drop and LinuxCNC shows estop.
- Check cabinet filters/fans; clean PC intake.

Monthly:

- Backup the LinuxCNC config directory and `tool.tbl` (copy to the GitHub repo + OneDrive).
- Inspect terminal strips for loosened screws (thermal cycling).
- Check drive fault history (TRA LED history where available; LinuxCNC logs).
- Test limit switches by slow jog toward each; confirm halt.
- Verify following-error trends per axis haven't degraded (halscope snapshot).

Chapter 3 — Lubrication & Hydraulics

- Follow 413m033 for oil types, quantities, and intervals — the retrofit changed nothing mechanical.
- The lube pump is now controlled via 7i84U OUT12 (`lube-on`); low level appears on IN25. If lube logic misbehaves, verify the HAL timer settings against the original ladder behavior (YM2V39L) before suspecting the pump.
- Hydraulics power the ATC, tool clamp, gear shift, and (via N1J-L2-201) the Z brake circuit. Pressure OK is monitored by the Sanwa SPS-8T-PC-20 on IN27. **No ATC or gear operation should ever be commanded without pressure OK.**

Chapter 4 — Servo Drives & Resolver Feedback Health

- **TRA drives:** keep vent paths clear; record any fault LED codes with a photo before resetting. Drive faults land on 7i84U IN10–IN12.
- **Tachometers:** the drives close their velocity loop on tachs — brush/commutator wear shows as speed instability that no LinuxCNC tuning will fix.
- **Resolvers:** sealed and robust, but connectors (MS-type circular shells) and cabling are the weak point. Symptoms of feedback trouble: position noise at standstill, following errors that scale with speed, or an axis that "hunts."
- Annually (or after any cabinet work): re-scope resolver sin/cos return amplitude and compare to the bring-up baseline recorded in Vol 3 §5.

Chapter 5 — Mesa Board & LinuxCNC Health

- Keep the 7i97T/7i49/7i84U boards free of chip dust and coolant mist; verify cabinet sealing.
- After any LinuxCNC update: re-run the latency test, and diff the HAL pin dump against `mesa_hal_pins.txt` before trusting the config.
- Keep firmware/bitfile versions recorded in `mesa/mesa_firmware_checklist.md` ; never flash new firmware mid-project without re-verifying pin names.
- Keep a bootable clone or image of the control PC drive — the PC is now a machine spare part.

Chapter 6 — Troubleshooting — Motion

Symptom	Checks (in order)
Joint following error on enable	Drive enabled? S-ON wiring TB3; drive fault input; resolver counts moving with hand rotation?
Following error at speed	Resolver return amplitude; tach health; drive gain; ferror settings vs actual tuning

Axis runs away on enable	Stop. Analog polarity reversed — re-verify Vol 3 §8 step 4 before re-enabling
Position noise at standstill	Resolver shield/ground; cable routing near power; return amplitude too low
Wrong direction on jog	Reverse resolver scale sign AND verify limits still protect correct ends
Homing goes wrong way	Home switch wiring (TB5.7–9), HOME_SEARCH_VEL sign in INI
Axis dead, no fault	Enable chain, TB3 continuity, drive power, contactor state

Chapter 7 — Troubleshooting — Spindle

Symptom	Checks
Won't start	E-stop chain; gear confirm (IN17/IN18 — must have exactly one true); FR-SX fault (IN14); enable path (resolve TB3-vs-7i84U ownership per Vol 3 §8)
Wrong speed	AOUT3 scaling; FR-SX command mode (0–10 V vs bipolar); gear range vs commanded S
No at-speed signal	IN13 wiring; FR-SX speed-reach parameter
Trips under load	Gear range too high for cut; FR-SX fault code on drive display
Won't orient	Orient logic depends on spindle feedback — verify encoder.03 reservation status and orient hardware per 41434WB

Chapter 8 — Troubleshooting — ATC / Magazine

Golden rule: never jog axes to "free" a stalled ATC. Identify the stalled step first.

Symptom	Checks
M6 refused	Tool-avail inputs (IN2/IN3); zone confirms PRS-55/66 (IN0/IN1); hydraulic OK (IN27); orient complete
Unclamp fails	SOL-10 via RLY-3/4 energizing? PRS-8/9 confirm switches; hydraulic pressure; dual-coil vs single-valve behavior (Vol 4 §9 note)
Magazine wrong pocket	BCD inputs IN19–23 vs physical pocket — test decode with bcd2s watch; check PRS-21...25 alignment
Arm stalls mid-swap	Which interlock dropped? Record all IN0–IN3, IN15/IN16 states before touching anything
Gear/ATC solenoid chatter	Coil voltage sag; relay contact rating; suppression missing

Recovery: restore hydraulic pressure, resolve the blocking input, and complete the cycle in the direction it was moving where possible. The original ladder (YM2V39L) documents the intended sequence — consult it

before manual intervention.

Chapter 9 — Troubleshooting — Mesa Comms / HAL

Symptom	Checks
LinuxCNC won't start / hm2_eth error	ping 192.168.1.121; NIC static IP; cable; mesaflash --readhmid
Random real-time delays	Latency test; disable CPU power management; NIC IRQ sharing
7i84U pins missing	Smart-serial link (Vol 4 §5) continuity; TB4 wiring; sserial error counters in HAL
Input never changes	Field power bank A/B present? Correct bank for that input number? Sensor 24 V supply
Output won't fire	Watch the HAL pin first (halcmd show pin); then bank power; then relay coil
Everything dead after config edit	Diff against the repo copy — restore from Git

Chapter 10 — Spare Parts & Sourcing

- **Mechanical:** source from the Parts List 413LE02A000 breakdown; Mazak can quote by publication + serial number.
- **Drives/motors:** Mitsubishi TRA / HD81-12S parts are obsolete — used-market items. The sister build (srdco/MazakVQC1540) community is a practical sourcing lead.
- **Mesa boards:** keep a spare of each board type once in production; store the exact firmware/bitfile with it.
- **Recommended shelf spares:** interposing relays (RLY-1...7 type), 24 V supply (HR-11F-24 equivalent), prox switch equivalents for PRS types, fuses per branch list, control PC image on USB.

End of Volume 5. Continue to Volume 6 — Reference, Glossary & Index.